

Item	Result
Analyze	Pass (v4, ST Edge AI Core 2.2.0, STM32H7)
Output shape	128 ✓
Flash	1.60 MiB of 2 MiB (weights 1.53 MiB + runtime ~70 KiB)
RAM	1.16 MiB (activations 1.11 MiB + runtime 45 KiB). More than the 1 MiB internal RAM, so float32 needs SDRAM
Desktop validation	C model matches the original except at the two edge outputs (replicate-padding mismatch in the head)
MACs	tool: 9.95 G. My calculation: about 60 M. Real timing pending on target
PC speed	114 ms per sample (laptop, not the board)
Decision	Conditional go. INT8 or SDRAM is required

- **Date, team, board, tool versions:** ST Edge AI Core 2.2.0, X-CUBE-AI 10.2.0, STM32H747I-DISCO.
- **Board test:** the board runs and sends text over UART at 115200 baud.
- **Problems found and fixed:** (1) Conv1d with dilation failed with the “H not found” error, so the model was changed to 4D Conv2d; (2) a Slice was ignored (129 vs 128 outputs), so a “pick” layer was added; (3) replicate padding mismatch at the edge outputs, v5 pending.
- **MAC formula**, so it can be checked: per attention layer $3 \cdot 129 \cdot 96^2 + 2 \cdot 8 \cdot 129^2 \cdot 12 + 129 \cdot 96^2 + 2 \cdot 129 \cdot 96 \cdot 384 = 17,461,440$; three layers $\approx 52.4\text{M}$; with the conv layers $\approx 58.4\text{M}$ (v2) or $\approx 60.0\text{M}$ (v4, which has the extra pick layer).
- **The two memory options as estimates:** float32 needs the external SDRAM (on this board it is at address 0xD0000000, according to ST’s board driver). INT8 should be about a quarter of the size, which I estimate fits internal RAM. Label both as estimates.